Conditional Probability

And the odds ratio and risk ratio as conditional probability

Today’s lecture

- Probability trees
- Statistical independence
- Joint probability
- Conditional probability
- Marginal probability
- Bayes’ Rule
- Risk ratio
- Odds ratio

Probability example

- *Sample space*: the set of all possible outcomes.

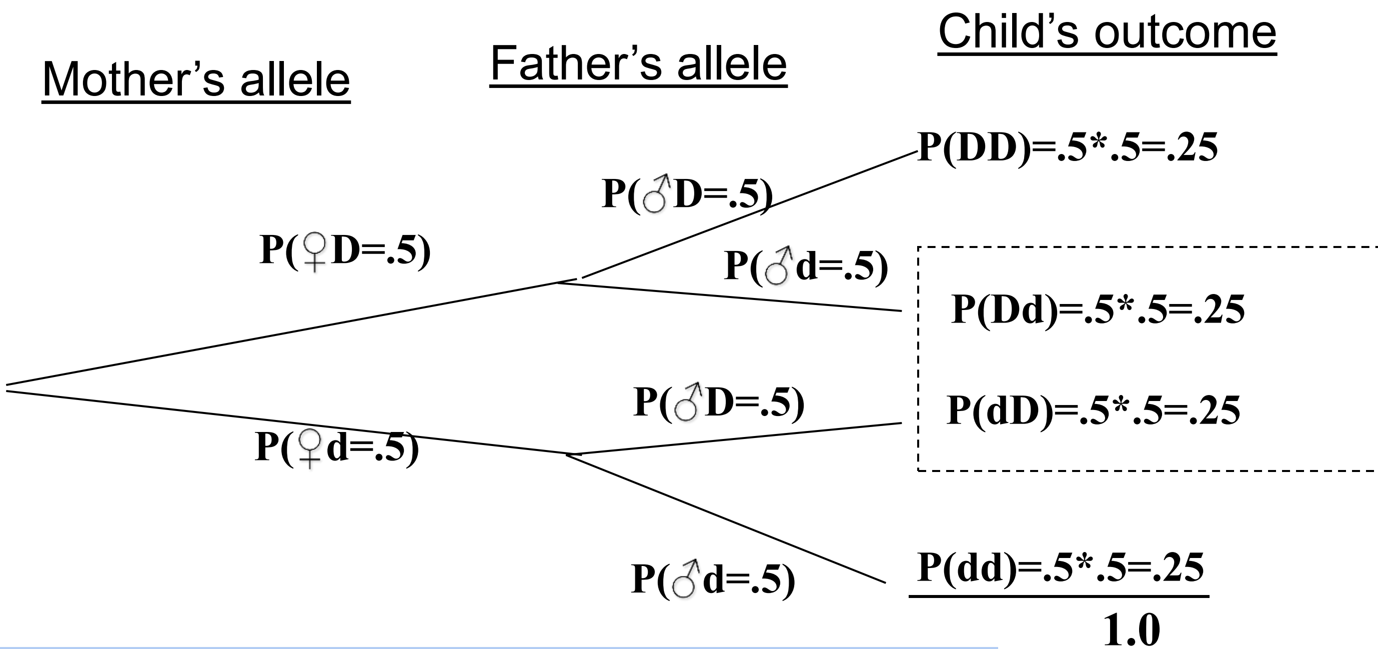
  For example, in genetics, if both the mother and father carry one copy of a recessive disease-causing mutation (d), there are three possible outcomes (the sample space):
  - child is not a carrier (DD)
  - child is a carrier (Dd)
  - child has the disease (dd).

- *Probabilities*: the likelihood of each of the possible outcomes (always $0 \leq P \leq 1.0$).
  - P(genotype=DD)=.25
  - P(genotype=Dd)=.50
  - P(genotype=dd)=.25.

Note: mutually exclusive, exhaustive probabilities sum to 1.

Using a probability tree

<u>Mendel example:</u> What's the chance of having a heterozygote child (Dd) if both parents are heterozygote (Dd)?



Rule of thumb: in probability, "and" means multiply, "or" means add

Independence

Formal definition: A and B are independent if and only if P(A&B)=P(A)*P(B)

The mother's and father's alleles are segregating independently.

P(♂D/♀D)=.5 and P(♂D/♀d)=.5

Conditional Probability: Read as "the probability that the father passes a D allele **given that** the mother passes a d allele."

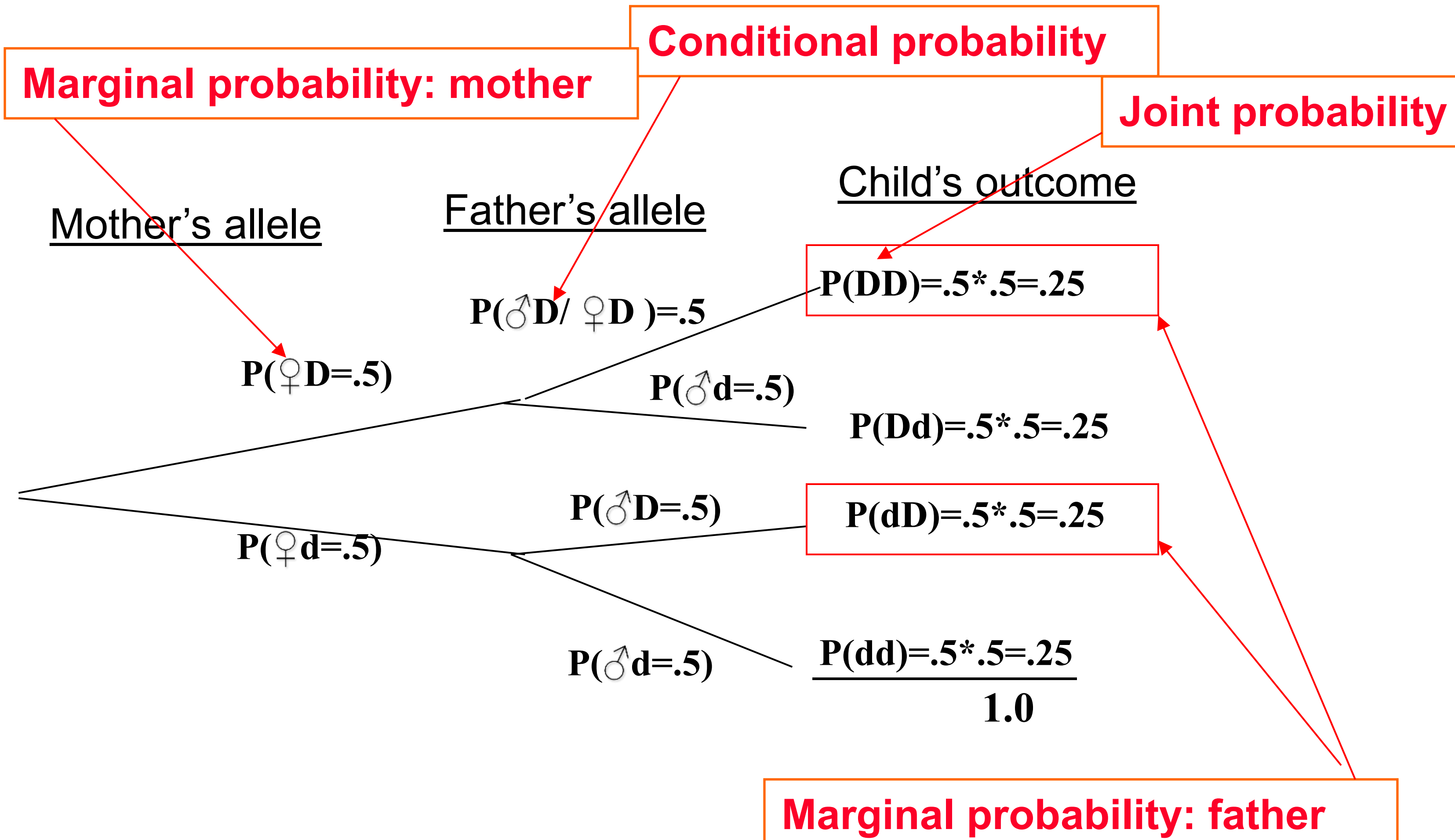
Joint Probability: The probability of two events happening simultaneously.

What father's gamete looks like is not dependent on the mother's – doesn't depend which branch you start on!

Marginal probability: This is the probability that an event happens at all, ignoring all other outcomes.

Formally, P(DD)=.25=P(D♂)*P(D♀)

On the tree


Conditional probability
Marginal probability: mother
Joint probability
Mother's allele
Father's allele
Child's outcome
P(DD)=.5*.5=.25
P(♂D/ ♀D)=.5
P(♀D=.5)
P(♂d=.5)
P(Dd)=.5*.5=.25
P(♂D=.5)
P(dD)=.5*.5=.25
P(♀d=.5)
P(♂d=.5)
P(dd)=.5*.5=.25
1.0
Marginal probability: father

Conditional, marginal, joint

- The marginal probability that player 1 gets two aces is 12/2652.
- The marginal probability that player 5 gets two aces is 12/2652.
- The marginal probability that player 9 gets two aces is 12/2652.
- The joint probability that all three players get pairs of aces is 0.
- The conditional probability that player 5 gets two aces given that player 1 got 2 aces is (2/50*1/49).

Test of independence

event A=player 1 gets pair of aces

event B=player 2 gets pair of aces

event C=player 3 gets pair of aces

- $P(A\&B\&C) = 0$
- $P(A)*P(B)*P(C) = (12/2652)^3$
- $(12/2652)^3 \neq 0$
- $\therefore$ Not independent

Independent $\neq$ mutually exclusive

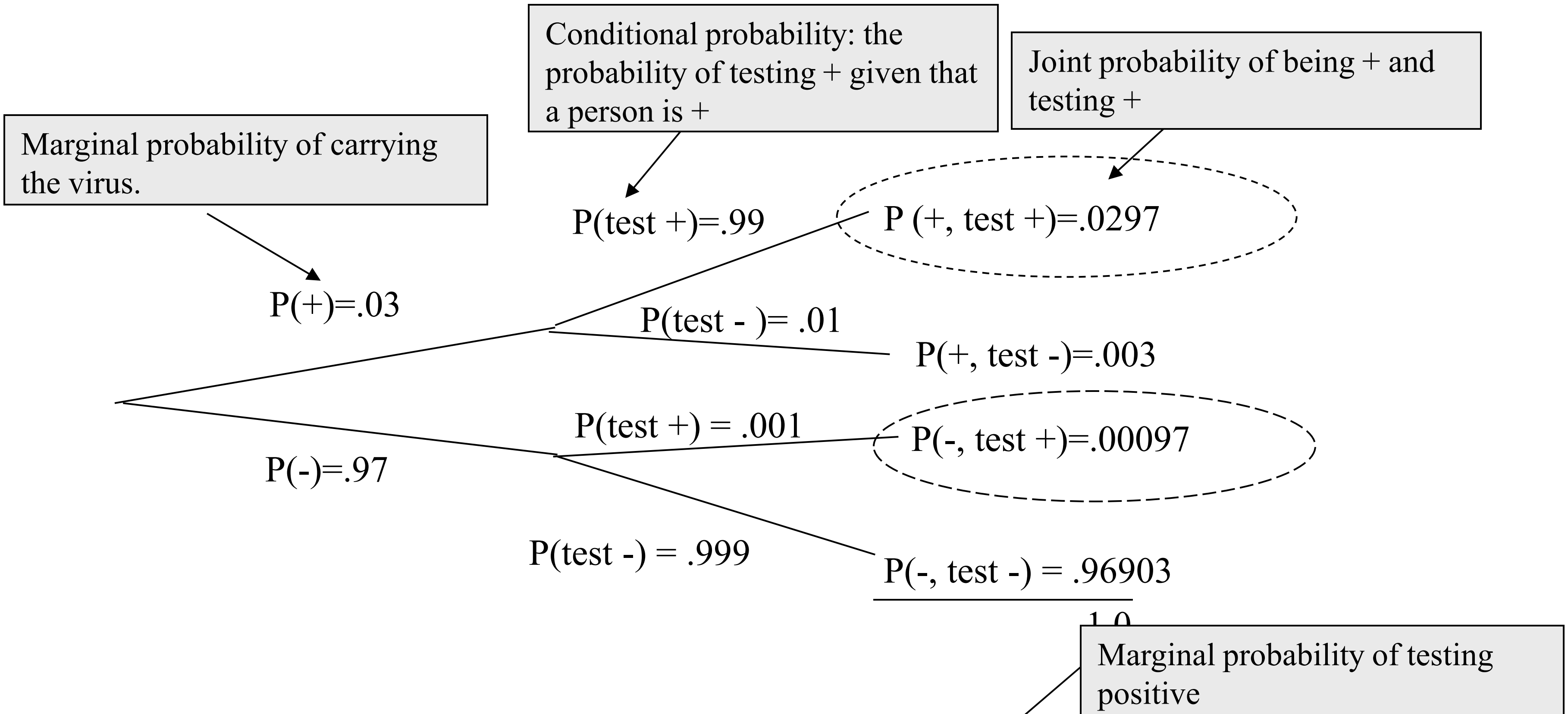
- Events A and ~A are mutually exclusive, but they are NOT independent.
- $P(A \& \sim A) = 0$
- $P(A) * P(\sim A) \neq 0$

Conceptually, once A has happened, ~A is impossible; thus, they are completely dependent.

Practice problem

If HIV has a prevalence of 3% in San Francisco, and a particular HIV test has a false positive rate of .001 and a false negative rate of .01, what is the probability that a random person selected off the street will test positive?

Answer



$$\therefore P(\text{test } +)=.0297+.00097=.03067$$

$P(+\&\text{test}+) \neq P(+)*P(\text{test}+)$

$.0297 \neq .03*.03067\ (=.00092)$

$\therefore$ Dependent!

Law of total probability

$$P(\text{test}+) = P(\text{test}+/\text{HIV}+)P(\text{HIV}+) + P(\text{test}+/\text{HIV}-)P(\text{HIV-})$$

One of these has to be true (mutually exclusive, collectively exhaustive). They sum to 1.0.

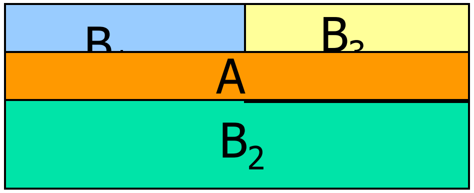
$$P(\text{test}+) = .99(.03) + .001(.97)$$

Law of total probability

- Formal Rule: Marginal probability for event A=

$$P(A) = P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2) + \cdots + P(A \mid B_k)P(B_k)$$

- Where: $\sum_{i=1}^{k} B_i = 1.0$ and $P(B_i \& B_j) = 0$ (mutually exclusive)

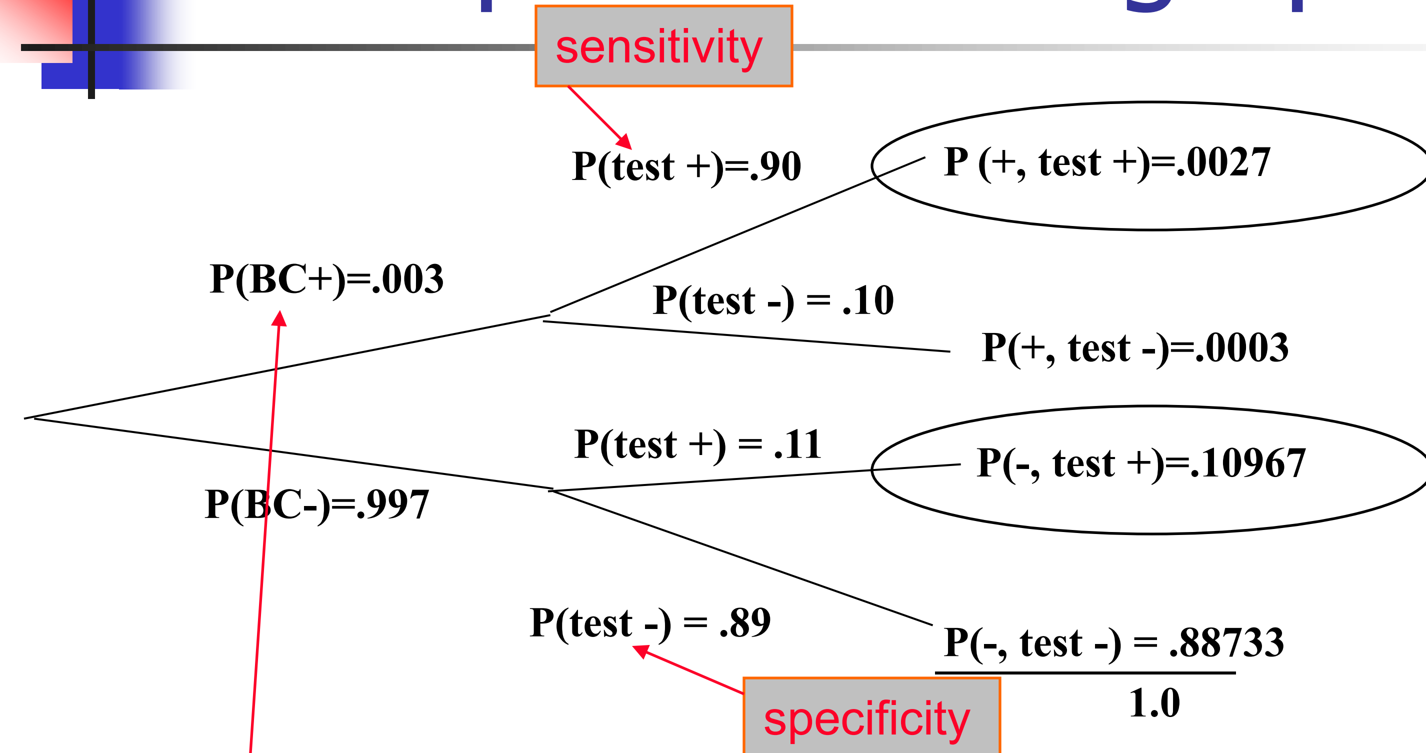


$$P(A) = (50\%)(25\%) + (0)(50\%) + \cdots + (50\%)(25\%) = 25\%$$

Example 2

- A 54-year old woman has an abnormal mammogram; what is the chance that she has breast cancer?

Example: Mammography



Marginal probabilities of breast cancer….(prevalence among all 54-

P(BC/test+)=.0027/(.0027+.10967)=2.4%

Bayes’ rule

Bayes' Rule: derivation

- <u>Definition:</u>

  Let A and B be two events with P(B) $\neq$ 0. The conditional probability of A given B is:

$$P(A/B) = \frac{P(A \& B)}{P(B)}$$

The idea: if we are given that the event B occurred, the relevant sample space is reduced to B {P(B)=1 because we know B is true} and conditional probability becomes a probability measure on B.

Bayes' Rule: derivation

$$P(A/B) = \frac{P(A\&B)}{P(B)}$$

can be re-arranged to:

$$P(A\&B) = P(A/B)P(B)$$

and, since also:

$$P(B/A) = \frac{P(A\&B)}{P(A)} \qquad \therefore P(A\&B) = P(B/A)P(A)$$

$$P(A/B)P(B) = P(A\&B) = P(B/A)P(A)$$

$$P(A/B)P(B) = P(B/A)P(A)$$

$$\therefore P(A/B) = \frac{P(B/A)P(A)}{P(B)}$$

Bayes' Rule:

$$P(A/B) = \frac{P(B/A)P(A)}{P(B)}$$

OR

$$P(A/B) = \frac{P(B/A)P(A)}{P(B/A)P(A) + P(B/\sim A)P(\sim A)}$$

From the "Law of Total Probability"

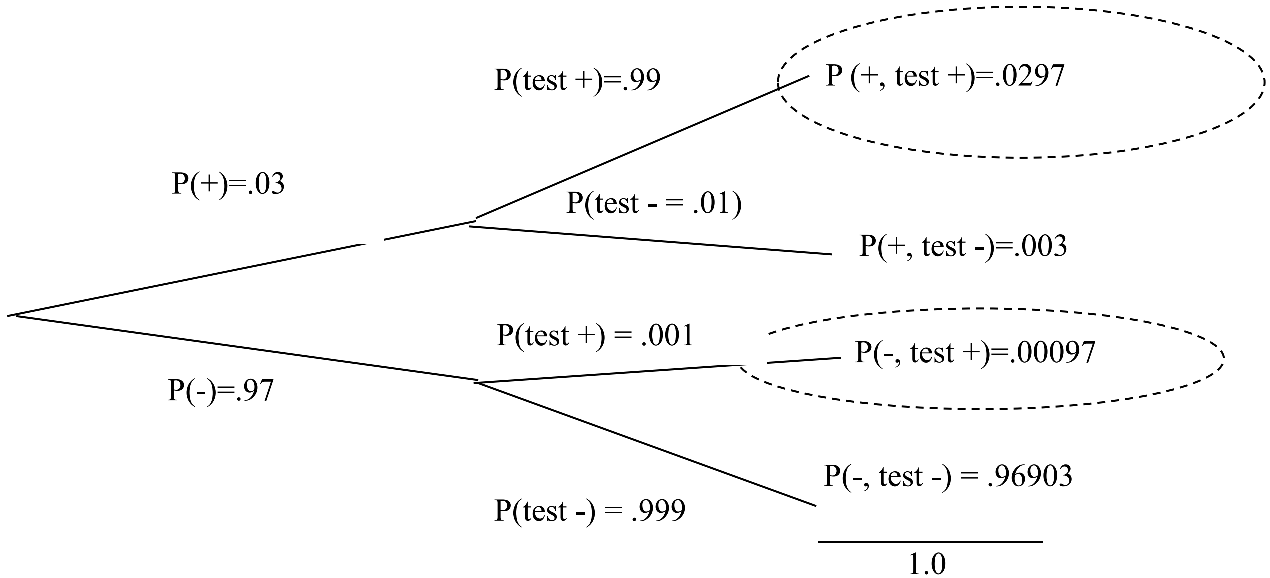
Bayes' Rule:

- Why do we care??
- Why is Bayes' Rule useful??
- It turns out that sometimes it is very useful to be able to "flip" conditional probabilities. That is, we may know the probability of A given B, but the probability of B given A may not be obvious. An example will help…

In-Class Exercise

- If HIV has a prevalence of 3% in San Francisco, and a particular HIV test has a false positive rate of .001 and a false negative rate of .01, what is the probability that a random person who tests positive is actually infected (also known as "positive predictive value")?

Answer: using probability tree



A positive test places one on either of the two "test +" branches.
But only the top branch also fulfills the event "true infection."
Therefore, the probability of being infected is the probability of being on the top branch given that you are on one of the two circled branches above.

$$P(+/test+) = \frac{P(test+\&true+)}{P(test+)} = \frac{.0297}{.0297+.00097} = 96.8\%$$

Answer: using Bayes' rule

$$P(true+/test+) = \frac{P(test+/true+)P(true+)}{P(test+/true+)P(true+) + P(test+/true-)P(true-)} =$$

$$\frac{.99(.03)}{.99(.03) + .001(.97)} = 96.8\%$$

Practice problem

An insurance company believes that drivers can be divided into two classes—those that are of high risk and those that are of low risk. Their statistics show that a high-risk driver will have an accident at some time within a year with probability .4, but this probability is only .1 for low risk drivers.

a) Assuming that 20% of the drivers are high-risk, what is the probability that a new policy holder will have an accident within a year of purchasing a policy?
b) If a new policy holder has an accident within a year of purchasing a policy, what is the probability that he is a high-risk type driver?

Answer to (a)

Assuming that 20% of the drivers are of high-risk, what is the probability that a new policy holder will have an accident within a year of purchasing a policy?

Use law of total probability:

P(accident)=

P(accident/high risk)*P(high risk) +

P(accident/low risk)*P(low risk) =

.40(.20) + .10(.80) = .08 + .08 = .16

Answer to (b)

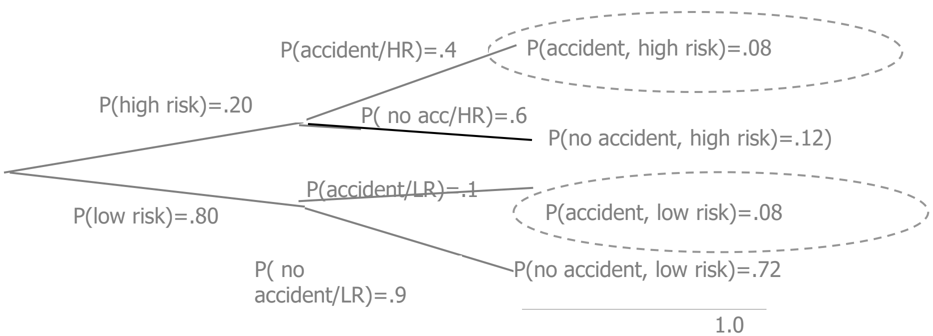
If a new policy holder has an accident within a year of purchasing a policy, what is the probability that he is a high-risk type driver?

P(high-risk/accident)=

P(accident/high risk)*P(high risk)/P(accident)

=.40(.20)/.16 = 50%

Or use tree:



P(high risk/accident)=.08/.16=50%

Fun example/bad investment

- http://www.cellulitedx.com/

Conditional Probability for Epidemiology:

The odds ratio and risk ratio as conditional probability

The Risk Ratio and the Odds Ratio as conditional probability

In epidemiology, the association between a risk factor or protective factor (exposure) and a disease may be evaluated by the "risk ratio" (RR) or the "odds ratio" (OR).

Both are measures of "relative risk"—the general concept of comparing disease risks in exposed vs. unexposed individuals.

Odds and Risk (probability)

Definitions:

Risk = P(A) = cumulative probability (you specify the time period!)

For example, what's the probability that a person with a high sugar intake develops diabetes in 1 year, 5 years, or over a lifetime?

Odds = P(A)/P(~A)

For example, "the odds are 3 to 1 against a horse" means that the horse has a 25% probability of winning.

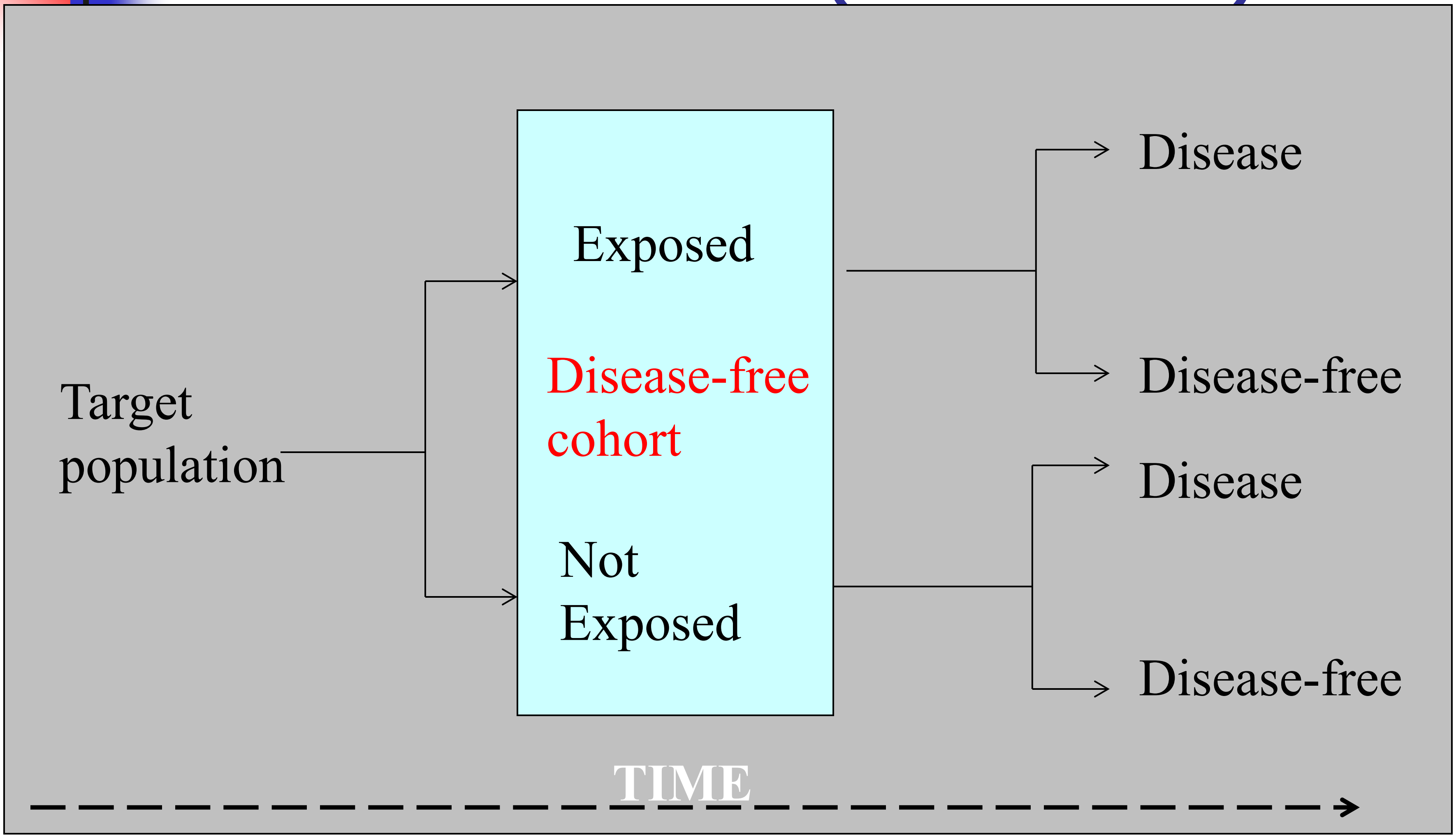
Note: An odds is always higher than its corresponding probability, unless the probability is 100%.

Odds vs. Risk=probability

If the risk is…	Then the odds are…
½ (50%)	**1:1**
¾ (75%)	**3:1**
1/10 (10%)	**1:9**
1/100 (1%)	**1:99**

Note: An odds is always higher than its corresponding probability, unless the probability is 100%.

Cohort Studies (risk ratio)



The Risk Ratio

	Exposure (E)	No Exposure (~E)
Disease (D)	a	b
No Disease (~D)	c	d
	a+c	b+d

risk to the exposed

$$RR = \frac{P(D/E)}{P(D/\sim E)} = \frac{\boxed{a/(a+c)}}{\boxed{b/(b+d)}}$$

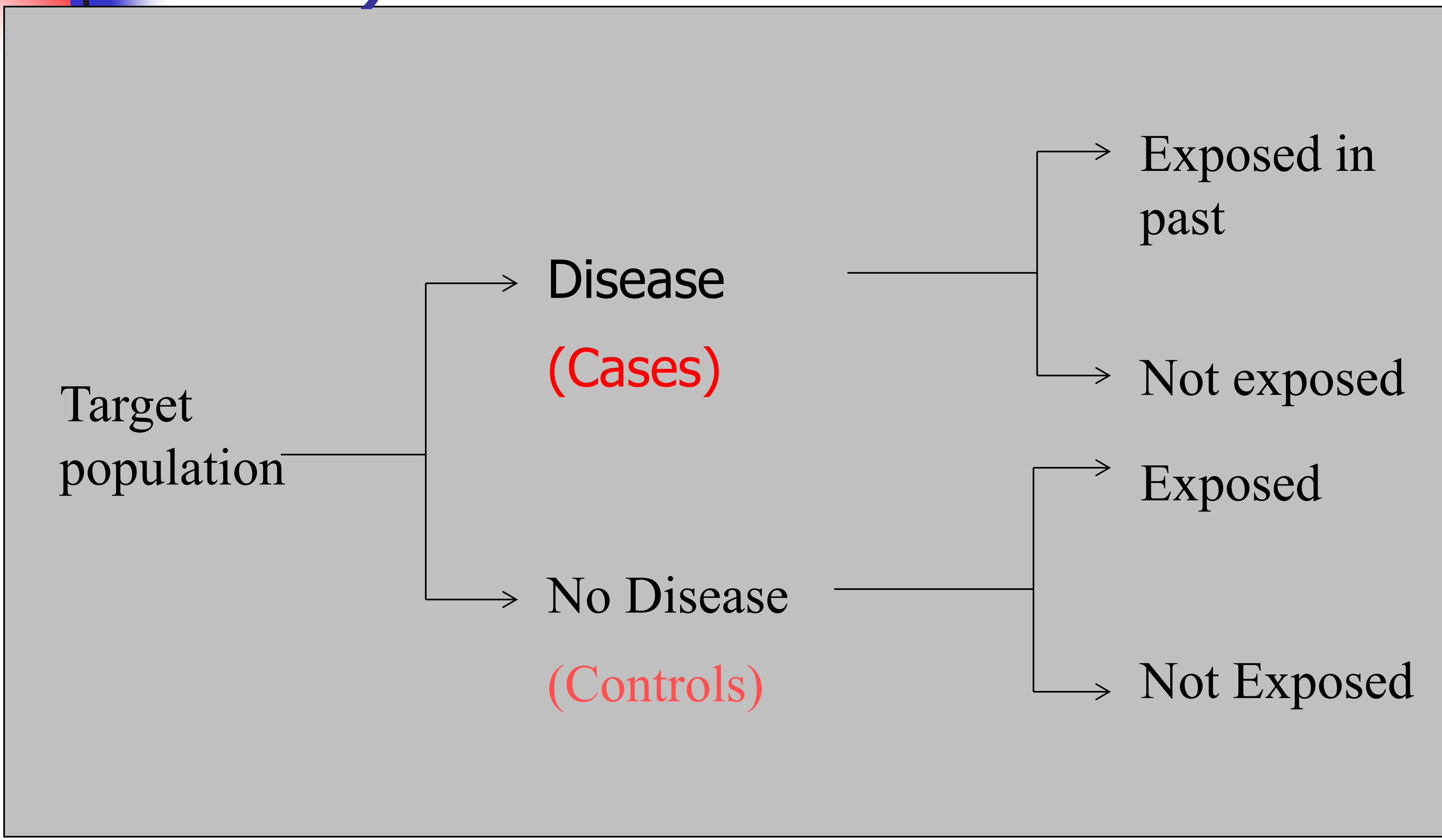
risk to the unexposed

Hypothetical Data

	High Systolic BP	Normal BP
Congestive Heart Failure	400	400
No CHF	1100	2600
	1500	3000

$$RR = \frac{400/1500}{400/3000} = 2.0$$

Case-Control Studies (odds ratio)



Case-control study example:

- You sample 50 stroke patients and 50 controls without stroke and ask about their smoking in the past.

Hypothetical results:

	Smoker (E)	Non-smoker (~E)	
Stroke (D)	**15**	**35**	50
No Stroke (~D)	**8**	**42**	50

What's the risk ratio here?

	Smoker (E)	Non-smoker (~E)	
Stroke (D)	**15**	**35**	50
No Stroke (~D)	**8**	**42**	50

Tricky: There is no risk ratio, because we cannot calculate the risk of disease!!

The odds ratio…

- We cannot calculate a risk ratio from a case-control study.

- BUT, we can calculate a measure called the odds ratio…

The Odds Ratio (OR)

	Smoker (E)	Smoker (~E)	
Stroke (D)	15	35	50
No Stroke (~D)	8	42	50

These data give: P(E/D) and P(E/~D).

Luckily, you can flip the conditional probabilities using Bayes' Rule:

$$P(D/E) = \frac{P(E/D)P(D)}{P(E)}$$

Unfortunately, our sampling scheme precludes calculation of the marginals: P(E) and P(D), but turns out we don't need these if we use an odds ratio because the marginals cancel out!

The Odds Ratio (OR)

	Exposure (E)	No Exposure (~E)
Disease (D)	a	b
No Disease (~D)	c	d

Odds of exposure in the cases

$$OR = \frac{\boxed{\dfrac{P(E/D)}{P(\sim E/D)}}}{\boxed{\dfrac{P(E/\sim D)}{P(\sim E/\sim D)}}} = \frac{\dfrac{a}{b}}{\dfrac{c}{d}} = \frac{ad}{bc}$$

Odds of exposure in the controls

The Odds Ratio (OR)

Odds of exposure in the cases

$$OR = \frac{\boxed{\dfrac{P(E/D)}{P(\sim E/D)}}}{\boxed{\dfrac{P(E/\sim D)}{P(\sim E/\sim D)}}}$$

Odds of exposure in the controls

But, this expression is mathematically equivalent to:

Odds of disease in the exposed

$$\frac{\boxed{\dfrac{P(D/E)}{P(\sim D/E)}}}{\boxed{\dfrac{P(D/\sim E)}{P(\sim D/\sim E)}}}$$

Odds of disease in the unexposed

Backward from what we want…

The direction of interest!

Proof via Bayes' Rule

$$\frac{\dfrac{P(E/D)}{P(\sim E/D)}}{\dfrac{P(E/\sim D)}{P(\sim E/\sim D)}}$$

Odds of exposure in the cases

Odds of exposure in the controls

Bayes' Rule

$$\dfrac{\dfrac{\dfrac{P(D/E)\cancel{P(E)}}{\cancel{P(D)}}}{\dfrac{P(D/\sim E)\cancel{P(\sim E)}}{\cancel{P(D)}}}}{\dfrac{\dfrac{P(\sim D/E)\cancel{P(E)}}{\cancel{P(\sim D)}}}{\dfrac{P(\sim D/\sim E)\cancel{P(\sim E)}}{\cancel{P(\sim D)}}}} =$$

$$\frac{\dfrac{P(D/E)}{P(\sim D/E)}}{\dfrac{P(D/\sim E)}{P(\sim D/\sim E)}}$$

Odds of disease in the exposed

Odds of disease in the unexposed

What we want!

The odds ratio here:

	Smoker (E)	Non-smoker (~E)	
Stroke (D)	**15**	**35**	50
No Stroke (~D)	**8**	**42**	50

$$OR = \frac{ad}{bc} = \frac{15 * 42}{35 * 8} = 2.25$$

- Interpretation: there is a 2.25-fold higher <u>odds</u> of stroke in smokers vs. non-smokers.

Interpretation of the odds ratio:

- The odds ratio will always be bigger than the corresponding risk ratio if RR >1 and smaller if RR <1 (the harmful or protective effect always appears larger)
- The magnitude of the inflation depends on the prevalence of the disease.

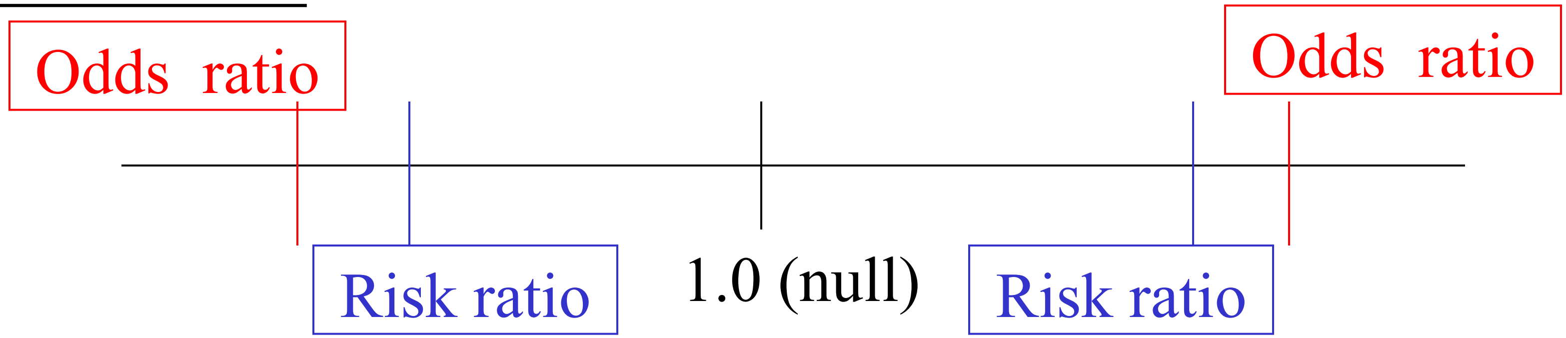
The rare disease assumption

$$OR = \frac{\dfrac{P(D/E)}{\cancel{P(\sim D/E)}_{\to 1}}}{\dfrac{P(D/\sim E)}{\cancel{P(\sim D/\sim E)}_{\to 1}}} \approx \frac{P(D/E)}{P(D/\sim E)} = RR$$

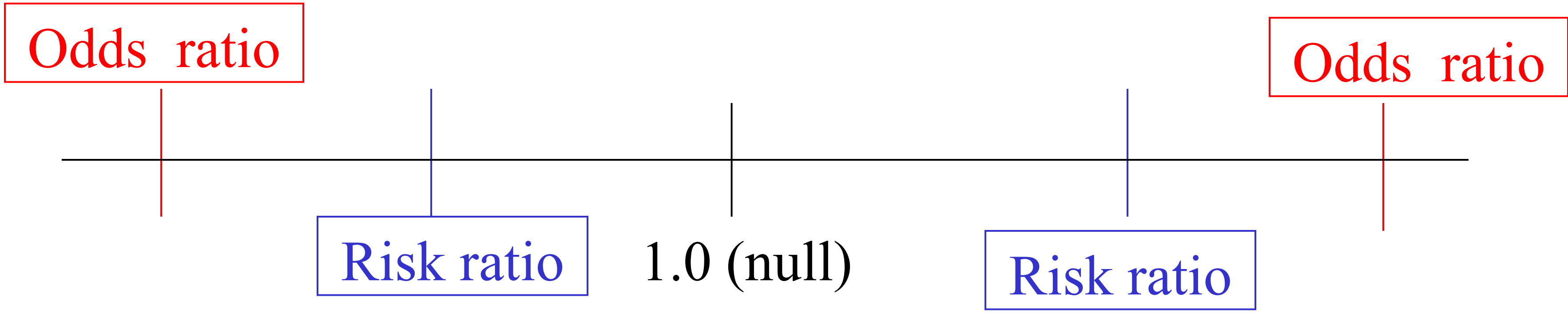
When a disease is rare:
P(~D) = 1 - P(D) ≅ 1

The odds ratio vs. the risk ratio

<u>Rare Outcome</u>



<u>Common Outcome</u>



Odds ratios in cross-sectional and cohort studies…

- Many cohort and cross-sectional studies report ORs rather than RRs even though the data necessary to calculate RRs are available. Why?
  - If you have a binary outcome and want to adjust for confounders, you have to use logistic regression.
  - Logistic regression gives adjusted odds ratios, not risk ratios (more on this in HRP 261).
  - These odds ratios must be interpreted cautiously (as increased odds, not risk) when the outcome is common.
  - When the outcome is common, authors should also report unadjusted risk ratios and/or use a simple formula to convert adjusted odds ratios back to adjusted risk ratios.

Example, wrinkle study…

- A cross-sectional study on risk factors for wrinkles found that heavy smoking significantly increases the risk of prominent wrinkles.
  - Adjusted OR=3.92 (heavy smokers vs. nonsmokers) calculated from logistic regression.
  - Interpretation: heavy smoking increases risk of prominent wrinkles nearly 4-fold??
  - The prevalence of prominent wrinkles in non-smokers is roughly 45%. So, it's not possible to have a 4-fold increase in risk (=180%)!

Raduan et al. *J Eur Acad Dermatol Venereol*. 2008 Jul 3.

Interpreting ORs when the outcome is common…

- If the outcome has a 10% prevalence in the unexposed/reference group*, the *maximum* possible RR=10.0.
- For 20% prevalence, the maximum possible RR=5.0
- For 30% prevalence, the maximum possible RR=3.3.
- For 40% prevalence, maximum possible RR=2.5.
- For 50% prevalence, maximum possible RR=2.0.

*Authors should report the prevalence/risk of the outcome in the unexposed/reference group, but they often don't. If this number is not given, you can usually estimate it from other data in the paper (or, if it's important enough, email the authors).

Interpreting ORs when the outcome is common…

If data are from a cross-sectional or cohort study, then you can convert ORs (from logistic regression) back to RRs with a simple formula:

$$RR = \frac{OR}{(1 - P_o) + (P_o \times OR)}$$

Where:

OR = odds ratio from logistic regression (e.g., 3.92)

P_0 = P(D/~E) = probability/prevalence of the outcome in the unexposed/reference group (e.g. ~45%)

Formula from: Zhang J. What's the Relative Risk? A Method of Correcting the Odds Ratio in Cohort Studies of Common Outcomes *JAMA.* 1998;280:1690-1691.

For wrinkle study…

$$RR_{\text{smokers vs. non-smokers}} = \frac{3.92}{(1-.45)+(.45\times 3.92)} = 1.69$$

So, the risk (prevalence) of wrinkles is increased by 69%, not 292%.

Zhang J. What's the Relative Risk? A Method of Correcting the Odds Ratio in Cohort Studies of Common Outcomes *JAMA.* 1998;280:1690-1691.

Sleep and hypertension study…

- $OR_{hypertension}$= 5.12 for chronic insomniacs who sleep ≤ 5 hours per night vs. the reference (good sleep) group.
- $OR_{hypertension}$ = 3.53 for chronic insomiacs who sleep 5-6 hours per night vs. the reference group.
- Interpretation: risk of hypertension is increased <u>500% and 350%</u> in these groups?
- No, ~25% of reference group has hypertension. Use formula to find corresponding RRs = 2.5, 2.2
- Correct interpretation: Hypertension is <u>increased 150% and 120%</u> in these groups.

-Sainani KL, Schmajuk G, Liu V. A Caution on Interpreting Odds Ratios. *SLEEP, Vol. 32, No. 8, 2009*.
-Vgontzas AN, Liao D, Bixler EO, Chrousos GP, Vela-Bueno A. Insomnia with objective short sleep duration is associated with a high risk for hypertension. Sleep 2009;32:491-7.

Practice problem:

1. Suppose the following data were collected on a random sample of subjects (the researchers did not sample on exposure or disease status).

	Neck pain	No Neck Pain
Own a cell phone	143	209
Don't own a cell phone	22	69

- Calculate the odds ratio and risk ratio for the association between cell phone usage and neck pain (common outcome).

Answer

	Neck pain	No Neck Pain
Own a cell phone	143	209
Don't own a cell phone	22	69

- OR = (69*143)/(22*209) = 2.15
- RR = (143/352)/(22/91) = 1.68

Practice problem:

- 2. Suppose the following data were collected on a random sample of subjects (the researchers did not sample on exposure or disease status).

	Brain tumor	No brain tumor
Own a cell phone	5	347
Don't own a cell phone	3	88

Calculate the odds ratio and risk ratio for the association between cell phone usage and brain tumor (rare outcome).

Answer

	Brain tumor	No brain tumor
Own a cell phone	5	347
Don't own a cell phone	3	88

- OR = (5*88)/(3*347) = .42267
- RR = (5/352)/(3/91) = .43087

Thought problem…

- Another classic first-year statistics problem. You are on the Monty Hall show. You are presented with 3 doors (A, B, C), only one of which has something valuable to you behind it (the others are bogus). You do not know what is behind any of the doors. You choose door A; Monty Hall opens door B and shows you that there is nothing behind it. Then he gives you the option of sticking with A or switching to C. Do you stay or switch? Does it matter?

Some Monty Hall links…

- http://query.nytimes.com/gst/fullpage.html?res=9D0CEFDD1E3FF932A15754C0A967958260&sec=&spon=&pagewanted=all
- http://www.nytimes.com/2008/04/08/science/08tier.html?_r=1&em&ex=1207972800&en=81bdecc33f60033e&ei=5087%0A&oref=slogin
- http://www.nytimes.com/2008/04/08/science/08monty.html#